Salifort Motors Employee Retention - Analysis

Executive Summary - Ben Cochrane - May 2026

Project Overview

Goal: Develop an effective statistical model for predicting employee retention and recommend strategies to improve company retention

Result: Logistic Regression, XGBoost, and Random Forest Models were trained and validated. Random Forest model performed best with an F1 score of 95.3% (Recall=92.5%, Precision=98.5%) on test data.

Key Insights

- Most important features were:

a) satisfaction level
b) number of projects
c) years employed
d) average monthly hours
e) last evaluation

- **Recall** of 92.5%

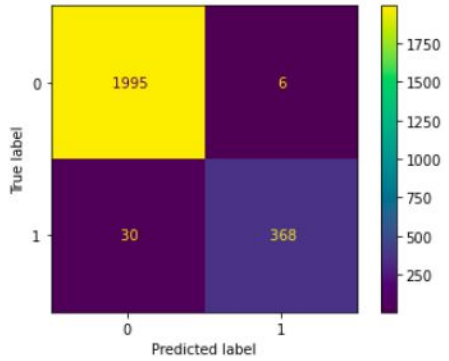
92.5% of people who actually left the company were correctly predicted to leave

- **Precision** of 98.4%

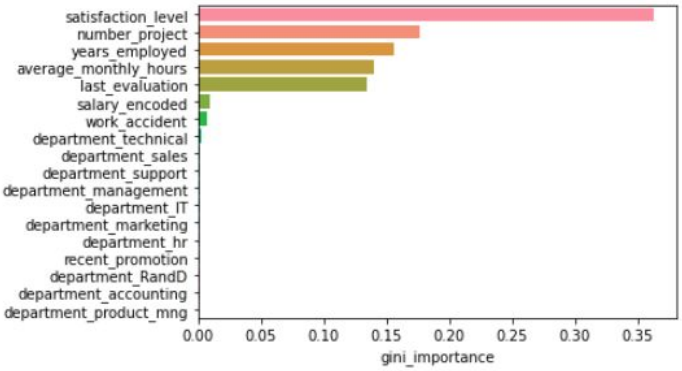
98.4% of the people the model predicted would leave did in fact leave the company.

- Employees with over **4 years of tenure** tended to be satisfied and not leave.
- Employees who had **six or seven projects** and were working large numbers of hours each month were very likely to leave.

Details



Random Forest Model - Test data Confusion Matrix



Random Forest Model - Test data feature importance

Next Steps

- Further questions about long tenured employees:
  - Did they have higher salaries? Did they have more seniority or better job titles? Why do they stay?
- Investigate ways to increase retention via addressing work hours and project numbers:
  - Project number caps
  - Project load sharing
  - Bonuses/rewards for long hours/many projects
- High average monthly hours worked and no variation in retention across departments suggests whole of organization work culture could be addressed
- Improve model further with feature engineering and data leakage considered